Dashboard Design

Date	24 September 2026
Team ID	
Project Name	EV Market Analytics Dashboard
Maximum Marks	5 Marks

Activity 1: Interactive and visually appealing dashboards

Creating an effective Streamlit dashboard involves presenting business data stored in Snowflake through clear, interactive, and easy-to-understand visualizations. Students should design the dashboard so that users can quickly understand important KPIs, trends, comparisons, and business insights.

- Clear and Intuitive Layout – KPI cards at the top followed by three pages: Executive Overview, Manufacturer & Market Share, Battery & Technology.
- Use Appropriate Visualizations – bar charts for categories, line chart for trends, donut chart for share, map for countries and scatter plot for capacity vs range.
- Colour and Theming – a consistent colour palette per battery type and vehicle category with a clean light theme.
- Interactive Filters and Slicers – sidebar filters for year, vehicle category, manufacturer, battery type and manufacturing country.
- Drill-Down Capabilities – from category to manufacturer to model through filters and tooltips.
- Responsive Design – wide layout with columns that resize on different screen sizes.
- Custom Visuals and Icons – KPI cards with icons and tooltips on all charts.
- Use of Infographics – summary KPI tiles and highlighted top-performer indicators.

[Insert Streamlit dashboard screenshot 1: Executive Overview]

[Insert Streamlit dashboard screenshot 2: Manufacturer & Battery pages]

Major outcomes:

1. **Sales concentration by vehicle category is visible at a glance**
   The Sales by Vehicle Category chart shows which segments (hatchback, sedan, SUV, bus, truck, two-wheeler) contribute most to annual sales, helping prioritise the product portfolio.
2. **Manufacturer market share highlights the competitive landscape**
   The market share and top manufacturer charts show which brands lead in sales and how their customer ratings compare.
3. **Battery technology adoption can be compared directly**
   The battery type charts show how LFP, NMC, NCA, Sodium-Ion and Solid-State batteries compare in adoption, driving range and capacity.

4. **Range and customer satisfaction can be compared across segments**
   Range by category and rating by category charts show which segments combine long range with high satisfaction.
5. **Manufacturing country analysis shows geographic production trends**
   The country map and chart show which countries lead EV production and sales, supporting market expansion decisions.

Overall outcome: The dashboard supports data-driven decisions on product portfolio, competitive positioning and battery technology strategy. [Replace the descriptions above with the exact values from your dashboard, e.g. the top category and manufacturer.]